INTERNAL ASSESSMENT ACTIVITY 3

AI-Assisted Literature Review, Digital Mapping and Comparative Analysis of Sequential and Foundation Models

Course : Advanced Neural Network Architectures (2307411T)
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Selected application: Short-Term Electrical Load Forecasting Based on Time Augmented Transformer (TAT)

Primary paper : Guangqi Zhang, Chuyuan Wei, Changfeng Jing and Yanxue Wang, "Short-Term Electrical Load Forecasting Based on Time Augmented Transformer," International Journal of Computational Intelligence Systems 15, Article 67 (2022). DOI: 10.1007/s44196-022-00128-y.

1. Problem Statement and Objectives :

Problem statement :

Given recent electricity-load observations and associated variables such as temperature, humidity, time-of-day and electricity price, predict future electrical load at short horizons (30 minutes, 1 hour, 12 hours and 1 day) with low error and practical computational cost.

Objectives :

1. Review the primary paper and explain TAT.
2. Describe the dataset, preprocessing, forecast horizons and metrics.
3. Compare the reported TAT results with ARIMA, SVR, LSTM, Bi-LSTM, CNN-LSTM, Seq2Seq and basic Transformer.
4. Map model evolution.
5. Identify limitations and propose future research.

2. Methodology of the Primary Paper :

Step	Component	Description
1	Input	48 historical half-hourly time steps containing load, weather, time, and price features.
2	Preprocessing	Zero-mean normalization and sliding-window generation of supervised samples.
3	Input Layer	Linear projection, positional encoding, and time-augmented encoding.
4	Encoder	Multi-head self-attention and feed-forward layers to learn relationships in the input sequence.
5	Decoder	Masked self-attention and encoder–decoder attention to generate predictions.
6	Linear Output Layer	Predicts electrical load 30 minutes, 1 hour, 12 hours, or 1 day ahead.

The authors formulate multi-step forecasting as supervised learning. A sliding window of historical observations predicts future load. Inputs include load and six other features: hour, dry-bulb temperature, wet-bulb temperature, dew-point temperature, humidity and electricity price. The data are from New South Wales, Australia, with half-hourly observations from 2006–2010.

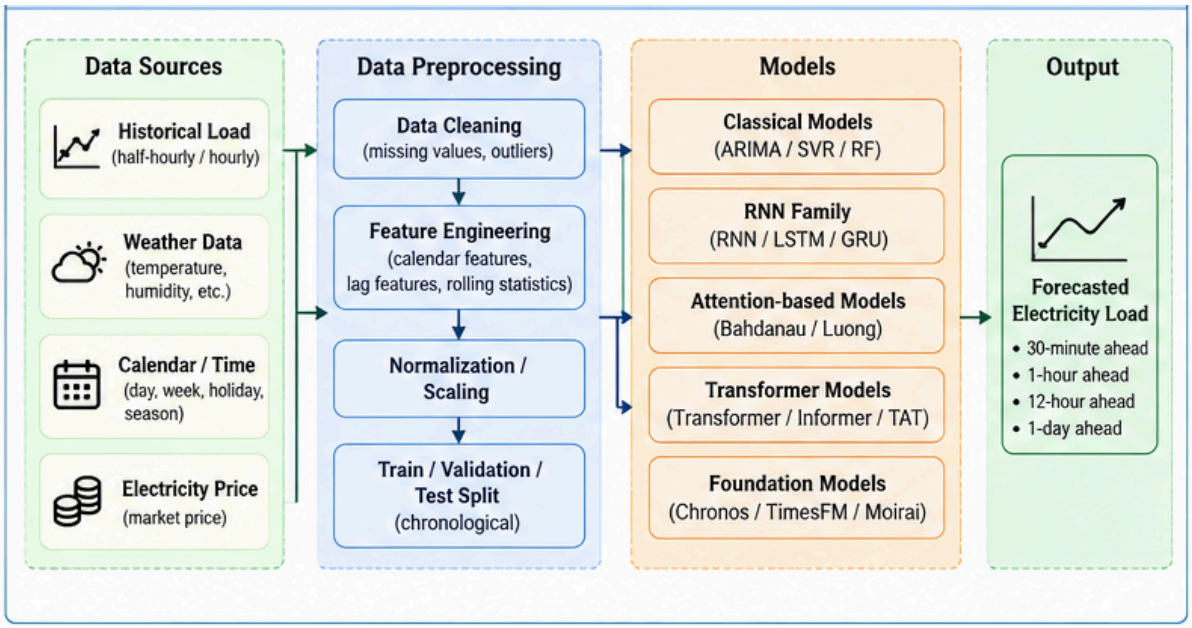


Figure 1. Overall Architecture for Electricity Load Forecasting

3. Results from the Primary Paper :

Multi-step Forecasting Comparison

Model	30m MAE	30m RMSE	30m MAPE	1h MAE	1h RMSE	1h MAPE	12h MAE	12h RMSE	12h MAPE	1d MAE	1d RMSE	1d MAPE
ARIMA	118.39	157.51	1.17 %	206.42	306.99	2.08 %	734.06	963.28	8.71%	800.57	1015.36	9.43%
SVR	86.55	109.71	1.01 %	197.91	238.67	2.37 %	374.90	439.61	4.51%	463.41	556.72	5.55%
LSTM	93.23	116.46	0.938 %	358.80	475.64	3.98 %	375.70	493.92	4.33%	498.85	649.25	5.82%
Seq2Seq	100.40	133.90	0.994 %	157.65	223.33	1.59 %	310.33	430.05	3.64%	486.88	639.13	5.68%
Bi-LSTM	83.36	106.09	0.858 %	117.65	156.66	1.20 %	314.21	447.06	3.68%	408.14	562.25	4.67%
CNN-LSTM	83.20	106.55	0.860 %	104.59	137.89	1.07 %	301.21	398.55	3.40%	370.51	483.21	4.24%
Transformer	78.81	99.91	0.744 %	84.19	113.38	0.82 %	281.10	376.69	3.27%	339.67	449.32	3.89%
TAT	47.20	62.49	0.465 %	65.99	87.59	0.618 %	147.55	215.18	1.69%	230.24	325.21	2.62%

Key observations :

- TAT performs best across all four forecasting horizons, achieving the lowest MAE, RMSE, and MAPE values in the supplied table.
- Transformer is the second-best overall model by the listed error metrics.
- Forecasting errors generally increase for longer prediction horizons.
- TAT achieves a MAPE of 0.465% at 30 minutes and 2.62% at one day, indicating better forecasting performance than the other listed models on these reported results.

4. AI-Assisted Literature Review :

ID / Year	Study	Category / Focus	Application / Data	Method and Contribution	Limitation / Caution
P1 (1997)	Hochreiter & Schmidhuber, “Long	LSTM foundation	General sequence learning	Gated memory cells address learning long dependencies and	Not a load-forecasting benchmark;

	Short-Term Memory”			vanishing-gradient issues in recurrent networks.	foundational method paper.
P2 (2018)	Tokgöz & Ünal, “A RNN Based Time Series Approach for Forecasting Turkish Electricity Load”	RNN; load forecasting	Turkish electricity load	Applies RNN-based time-series forecasting to electricity demand.	Results depend on the local dataset and setup; not directly comparable with TAT without a matching protocol.
P3 (2018)	Peng et al., “Short-Term Load Forecasting of Smart Grid Based on LSTM Recurrent Neural Networks…”	LSTM; load forecasting	Smart-grid load with real-time electricity-price context	Uses LSTM-RNN for short-term load forecasting with price context.	Publication and setup differ from TAT; exact numerical comparison requires the original paper.
P4 (2014/2015)	Bahdanau, Cho & Bengio, “Neural Machine Translation by Jointly Learning to Align and Translate”	Attention	Machine-translation sequence modelling	Introduces learned soft alignment/attention over input positions.	Not a load-forecasting study; provides the conceptual origin of attention.
P5 (2015)	Luong, Pham & Manning, “Effective Approaches to Attention-Based Neural Machine Translation”	Attention	Machine translation	Compares attention mechanisms for sequence-to-sequence learning.	Different domain and task; results should not be compared numerically with load forecasting.
P6 (2017)	Vaswani et al., “Attention Is All You Need”	Transformer	Sequence transduction	Introduces the Transformer with multi-head self-attention and positional encoding, removing recurrence from core sequence processing.	Not designed specifically for numeric load forecasting; temporal features may need adaptation.
P7 (2021)	Zhou et al., “Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting”	Efficient Transformer	Long-sequence forecasting benchmarks, including energy-related series	Uses ProbSparse attention, distilling and a generative decoder to improve efficiency for long sequences.	Benchmark datasets and protocol differ from the TAT study.

P8 (2022)	Zhang et al., “Short-Term Electrical Load Forecasting Based on Time Augmented Transformer”	TAT; primary paper	New South Wales NEM, half-hourly data, 2006–2010	Adds calendar-time augmentation to Transformer input and evaluates four forecast horizons against seven baselines.	One regional dataset; external validation and comparison with modern foundation models remain open.
P9 (2020)	Nti et al., “Electricity Load Forecasting: A Systematic Review”	Review	77 prior works across electricity-demand forecasting	Reviews methods, factors, forecast horizons and metrics used in load forecasting.	A synthesis rather than a new single-model benchmark; individual studies are heterogeneous.
P10 (2020)	Mamun et al., “A Comprehensive Review of the Load Forecasting Techniques Using Single and Hybrid Predictive Models”	Review	Electricity load forecasting	Reviews single and hybrid predictive approaches, motivating model selection by context.	Review conclusions do not substitute for a controlled local evaluation.
P11 (2024)	Ansari et al., “Chronos: Learning the Language of Time Series”	Time-series foundation model	42 benchmark datasets across domains	Scales and quantizes series into tokens and trains pretrained probabilistic Transformer models; reports zero-shot forecasting results.	Does not establish superiority on the TAT New South Wales split; domain fit and exogenous-variable support must be tested.
P12 (2024)	Das et al., “A Decoder-Only Foundation Model for Time-Series Forecasting” (TimesFM)	Time-series foundation model	Large corpus of real-world time points and public forecasting benchmarks	Decoder-only pretrained model aimed at zero-shot forecasting across domains and granularities.	General benchmark performance does not guarantee performance on regional electricity demand.

5. Evolution of Models :

Model stage	Innovation	Limitation	Motivation / Next step
RNN	Recurrent hidden state processes sequences in order.	Vanishing/exploding gradients; difficulty learning long-term dependencies.	LSTM/GRU add gated memory.
LSTM / GRU	Gates regulate what information is retained or forgotten.	Sequential computation; can still struggle with very long sequences.	Attention lets outputs focus on relevant information.

Attention	Query–key–value weighting selects relevant information.	Attention can be computationally expensive.	Transformer integrates attention as the core mechanism.
Transformer	Multi-head self-attention and positional encoding enable parallel sequence processing.	Standard attention can require substantial memory and computation.	Efficient variants and domain-specific temporal features improve performance.
Pretrained time-series models	Pretrain on diverse time series, enabling zero-shot or few-shot forecasting.	Distribution shift and covariate mismatch can reduce accuracy.	Foundation models aim to transfer knowledge across datasets and tasks.
Foundation models / LLMs	Large pretrained models support broad transfer across forecasting tasks.	Compute requirements, hallucinations, and domain mismatch.	Validate on the target task and combine forecasting with domain-specific information.

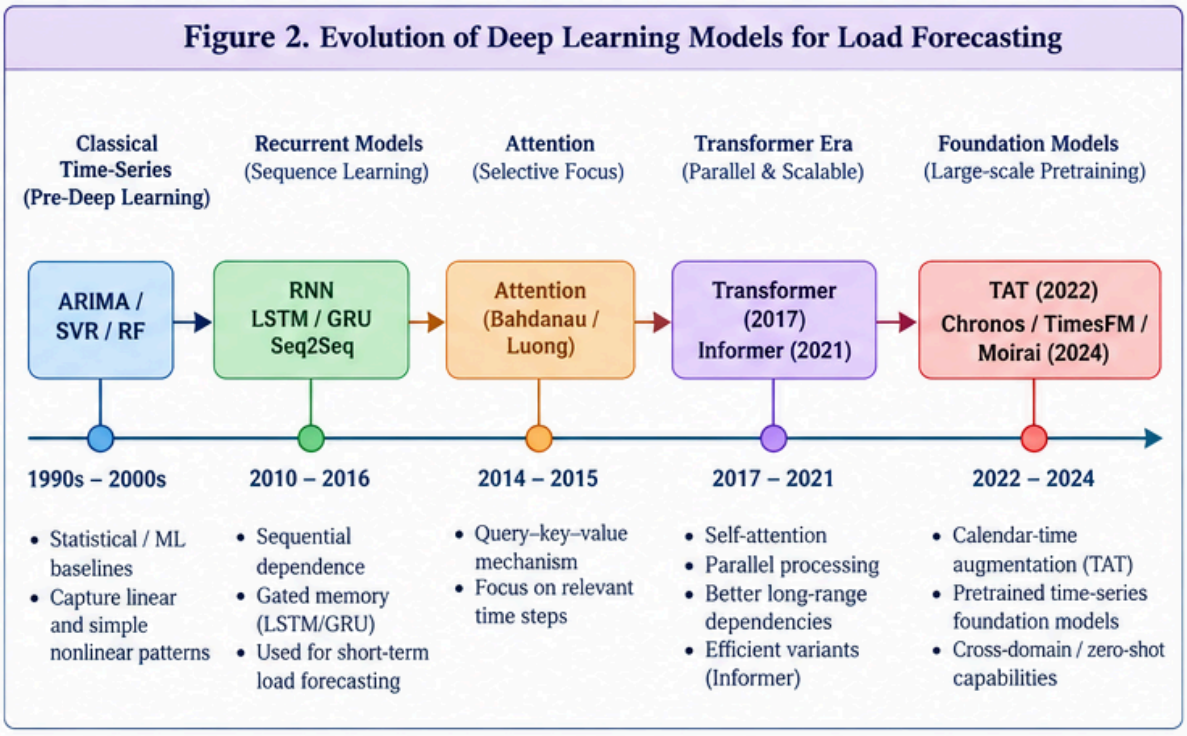


Figure 2. Evolution of Deep Learning Models

6. Digital Literature Map :

Step	Stage	Description
1	Problem	Forecast electricity demand 30 minutes to 1 day ahead.
2	Data layer	Historical load, weather, calendar/time features, and electricity price.
3	Classical baselines	ARIMA, SVR, Random Forest (RF), and boosting.
4	Model Families	RNN → LSTM / Bi-LSTM / GRU / Seq2Seq.
5	Evaluation	Chronological split, MAE / RMSE / MAPE, and external validation.
6	Key Finding	The primary paper reports that TAT achieves the lowest forecasting errors under the tested settings.

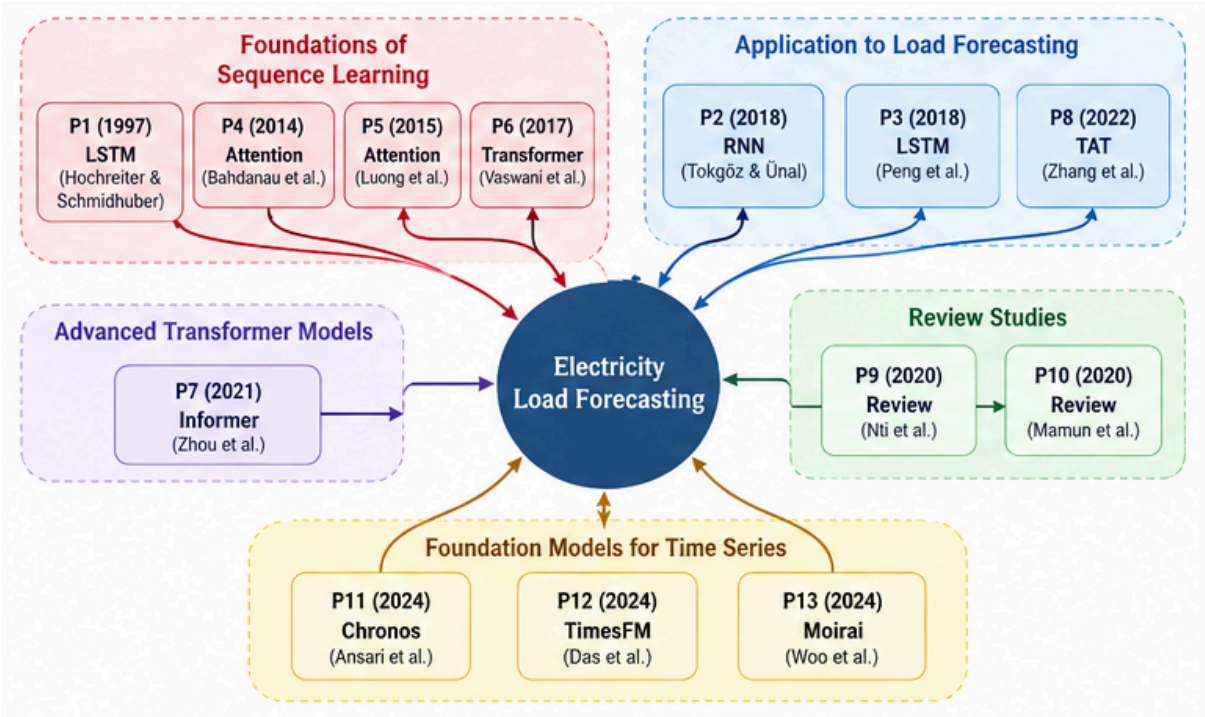


Figure 3. Digital Literature Map

7. Model-to-Application Mapping :

Model	Capability	Application in Load Forecasting	Advantage	Limitation / Caution
ARIMA	Autocorrelation and linear temporal patterns	Simple reference forecast	Interpretable and relatively lightweight	Stationarity and nonlinear-pattern limitations
SVR / Tree Models	Nonlinear relationships using engineered features	Forecast load from historical and exogenous features	Useful nonlinear baselines	Feature engineering and tuning affect results
RNN / LSTM / GRU	Sequential dependence and gated memory	Short-term load sequences	Can learn temporal patterns from data	Sequential computation; long dependencies may remain difficult
CNN-LSTM / Seq2Seq	Local patterns and sequence encoding/decoding	Multi-step demand prediction	Combines local and temporal information	Complexity; results depend on design and data
Transformer / Informer	Self-attention and long-range dependencies	Multivariate and long-horizon load forecasts	Parallel processing; flexible context integration	Attention cost; positional encoding alone may not capture calendar semantics
TAT	Transformer with explicit calendar-time augmentation	30-minute, 1-hour, 12-hour and 1-day load forecasts	Lowest reported errors among tested baselines in the primary experiment	One regional dataset; external validity not established
Chronos / TimesFM / Moirai	Pretrained time-series representations and zero-shot forecasting	Candidate for cross-region transfer or low-data deployment	May reduce per-dataset training needs	Must test covariates, calibration, distribution shift, runtime and performance on the same split
LLM as Operator Interface (Proposed)	Natural-language interaction and report generation	Explain forecasts, summarize anomalies and answer operator queries	Accessible interface to forecast outputs	Should not replace a validated numerical model; hallucination and traceability risks

8. Comparative Analysis :

Dimension	RNN	LSTM / GRU	Attention	Transformer / TAT	Foundation Models / LLMs
Sequential processing	Sequential recurrent updates	Sequential with gated state	Can be added to sequence models	Self-attention processes positions in parallel during training	Pretrained time-series models often use Transformer backbones; LLMs use token sequences
Long-term dependency	Can be difficult due to gradient issues	Gates help preserve information	Directly weights relevant positions	Self-attention connects distant time steps	Broad pretraining may help transfer, but domain fit is empirical
Parallel processing	Limited across time steps	Limited across time steps	Depends on backbone	Strong parallelism across sequence positions	Training can be compute-intensive; inference varies by model
Attention mechanism	Not inherent	Not inherent	Core mechanism	Core mechanism; TAT adds time encoding	Common in Transformer-based time-series models and LLMs
Scalability	Simple models can be lightweight	Moderate; recurrent bottleneck	Cost depends on length and design	Standard attention can have high memory/compute cost; efficient variants address this	Large pretraining cost; inference footprint may be significant
Data requirement	Can work with smaller datasets but has limited capacity	Often needs sufficient task data	Depends on architecture	Usually needs task data and careful tuning	Pretraining uses large corpora; zero-shot/few-shot use may reduce task-specific training
Application suitability	Baseline for sequence data	Established choice for load series	Useful when relevant time steps vary	Strong candidate for multivariate, longer-range forecasting	Candidate when transferring across regions/tasks; must be benchmarked
Major advantage	Simple sequential baseline	Gated memory for temporal patterns	Selective focus on relevant input	Global dependencies; TAT encodes calendar context	Reusable pretrained representations and possible zero-shot forecasts

9. Critical Analysis :

1. Why was LSTM developed after RNN?

Ans: Basic RNNs can struggle to learn long-range dependencies because gradients may vanish or explode during backpropagation through time. LSTM introduces a cell state and gates that regulate what is retained, added or forgotten, improving the ability to learn longer temporal relationships.

2. Why was Attention introduced?

Ans: A conventional encoder–decoder can compress an input sequence into a limited representation. Attention lets a model dynamically weight relevant input positions for each output, helping it use different parts of the history instead of relying only on one compressed vector.

3. What limitation of RNN/LSTM did Transformer address?

Ans: Transformers remove recurrent sequence updates from the core attention stack. Self-attention relates distant positions directly and permits parallel processing across input positions during training. This can improve long-range dependency learning and training efficiency, although standard attention can have high memory and computational cost.

4. Why are Transformers suitable for large-scale pretraining?

Ans: Their parallelizable sequence processing, scalable architecture and attention-based context representation support training on large datasets. However, scalability is not automatic: data quality, compute, tokenization/representation and objective design remain important.

5. Does a newer model always perform better?

Ans: No. Performance depends on the task, data, horizon, features, tuning and evaluation protocol. The TAT paper reports lower errors than its tested baselines, but it does not evaluate later foundation models. A simpler model may be preferable when data are scarce, latency is strict or interpretability is critical.

6. Which model is most suitable for the selected application?

Ans: For the exact New South Wales experiment in the primary paper, TAT is the strongest reported model because it has the lowest MAE, RMSE and MAPE at all four tested horizons. For deployment elsewhere, the recommendation is conditional: reproduce TAT, compare it with local baselines and test a pretrained time-series model on the same split before selecting a model.

7. What are the computational limitations?

Ans: RNN/LSTM process time steps sequentially, limiting parallelism. Standard Transformer attention can require substantial memory and compute as sequence length grows. Foundation models add pretraining and inference costs. The study should measure inference latency, peak memory and hardware needs, not only forecast errors.

8. What research gaps are identified?

Ans: The main gaps are external validation, direct comparison with modern time-series foundation models, robustness to missing/noisy features and distribution shift, calibrated uncertainty, interpretability, and evidence of operational value. These gaps follow from what is not established by the primary paper's reported experiments.

9. What future model/approach could address these gaps?

Ans: A fair cross-region benchmark should compare TAT, tuned statistical/recurrent baselines and at least one pretrained time-series foundation model (Chronos, TimesFM or Moirai). Use identical chronological splits and forecast horizons, report MAE/RMSE/MAPE plus uncertainty calibration and runtime, perform ablations on time/weather/price features, and test robustness under missing covariates and demand shifts. An LLM may summarize forecast changes for operators, but should not replace the validated numerical forecasting model.

9. Research Gap Map and Future Proposal :

Stage	Description
Evidence	TAT reports the lowest MAE, RMSE and MAPE among the tested baselines on New South Wales (NSW) data.
Gap 1: External validation	Validation across regions, years, seasons and market conditions is required.
Gap 2: Foundation-model comparison	No direct comparison with modern pretrained time-series foundation models is presented.
Gap 3: Forecast uncertainty	Point metrics alone do not quantify forecast uncertainty.
Gap 4: Feature reliability	Calendar, weather and electricity-price feature reliability, along with missing data, needs stress testing.
Gap 5: Interpretability and operational value	Interpretability and operational value are not established by error metrics alone.
Proposed research direction	Conduct a leakage-safe, cross-region benchmark comparing TAT with Chronos, TimesFM and Moirai.
Evaluation metrics	MAE, RMSE, MAPE, prediction intervals, latency, memory usage, ablation studies and drift tests.

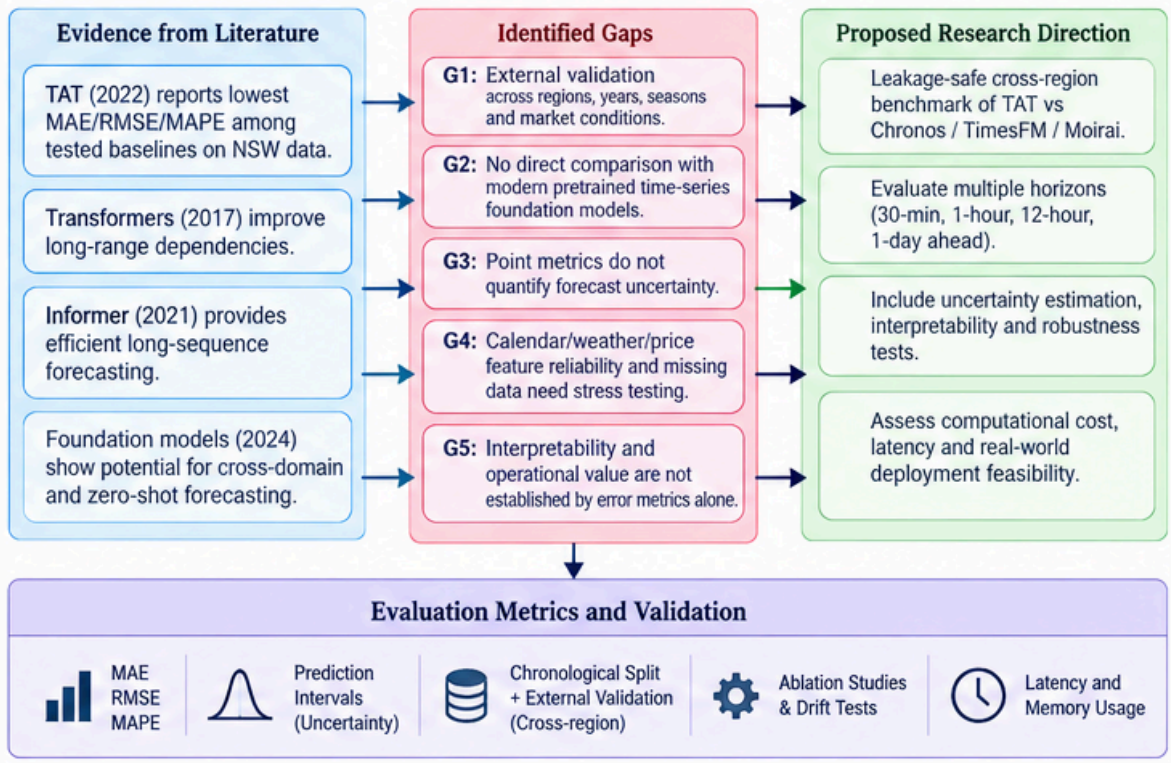


Figure 4. Research Gap Analysis and Future Direction

10. AI Usage and Verification Log :

Tool	Purpose / example query	Information used	Verification
ChatGPT	Concept explanation, report organization, comparison and draft synthesis	Draft structure, explanations and gap hypotheses	Check every factual claim against the supplied TAT paper or cited original publication; student must verify.
Google Scholar / Semantic Scholar	Discover and cross-check paper metadata and abstracts	Titles, author/year, venue and high-level contributions	Use DOI/publisher/arXiv/PMLR records; consult full text before final submission.
Scholarly web search / publisher pages	Discover and cross-check paper metadata and abstracts	Titles, author/year, venue and high-level contributions	Use DOI/publisher/arXiv/PMLR records; consult full text before final submission.

11. Conclusion :

The literature progresses from RNNs to attention, Transformers, and pretrained foundation models. The TAT study reports the lowest errors among tested baselines across four forecasting horizons, with multivariate inputs outperforming univariate inputs. However, it uses only one regional dataset and does not compare modern foundation models. Future research should focus on cross-region validation, uncertainty, robustness, and computational efficiency.

12. References :

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